

module4_lab

September 27, 2026

```
[1]: import pandas as pd
df = pd.read_csv("sales_clean.csv")
```

```
[2]: df.head()
```

```
[2]:
```

	order_id	date	product	price	qty	zip
0	1254	2026-04-06	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	2026-05-06	mug	5.50	1	10001
3	1066	2026-05-30	notebook	37.53	1	98101
4	1114	2026-04-06	webcam	45.59	4	90405

```
[3]: df.shape
```

```
[3]: (292, 6)
```

```
[4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 292 entries, 0 to 291
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   order_id    292 non-null    int64
 1   date        292 non-null    object
 2   product     292 non-null    object
 3   price       292 non-null    float64
 4   qty         292 non-null    int64
 5   zip         292 non-null    int64
dtypes: float64(1), int64(3), object(2)
memory usage: 13.8+ KB
```

```
[5]: df.describe()
```

```
[5]:
```

	order_id	price	qty	zip
count	292.000000	292.000000	292.000000	292.000000
mean	1145.500000	55.523151	2.910959	45463.989726
std	84.437354	71.767085	1.559457	39284.386854
min	1000.000000	2.930000	-5.000000	2116.000000

25%	1072.750000	11.045000	2.000000	2134.000000
50%	1145.500000	37.530000	3.000000	30303.000000
75%	1218.250000	55.510000	4.000000	90210.000000
max	1291.000000	479.290000	5.000000	98101.000000

```
[6]: print(df["price"].mean())
      print(df["price"].median())
      print(df["price"].std())
```

```
55.52315068493151
37.53
71.76708493504135
```

```
[ ]:
```

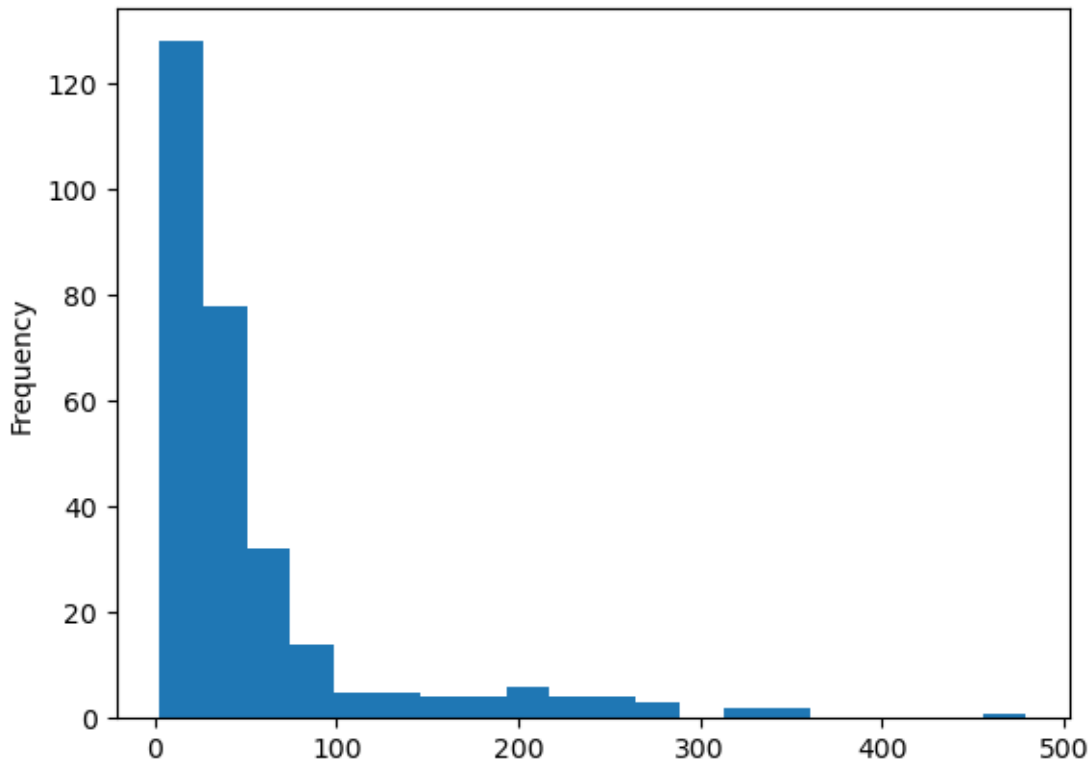
```
[7]: print(df["qty"].mean())
      print(df["qty"].median())
      print(df["qty"].std())
```

```
2.910958904109589
3.0
1.559457271336151
```

For price the mean is 55.52 and median is 37.53 The mean is noticeably higher than the median. I would report the median 37.53 because the mean is slightly skewed by the max price of 479.29.

```
[8]: df["price"].plot(kind="hist", bins=20)
```

```
[8]: <Axes: ylabel='Frequency'>
```



The shape of the histogram is right-skewed (positive) because most products are on the lower end, while a few expensive products create a long tail to the right.

```
[9]: df["revenue"] = df["price"] * df["qty"]
```

```
[10]: df.head()
```

```
[10]:
```

	order_id	date	product	price	qty	zip	revenue
0	1254	2026-04-06	mug	11.36	4	60614	45.44
1	1057	2026-05-30	charger	14.79	3	30303	44.37
2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	37.53	1	98101	37.53
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

```
[11]: df.corr(numeric_only=True)
```

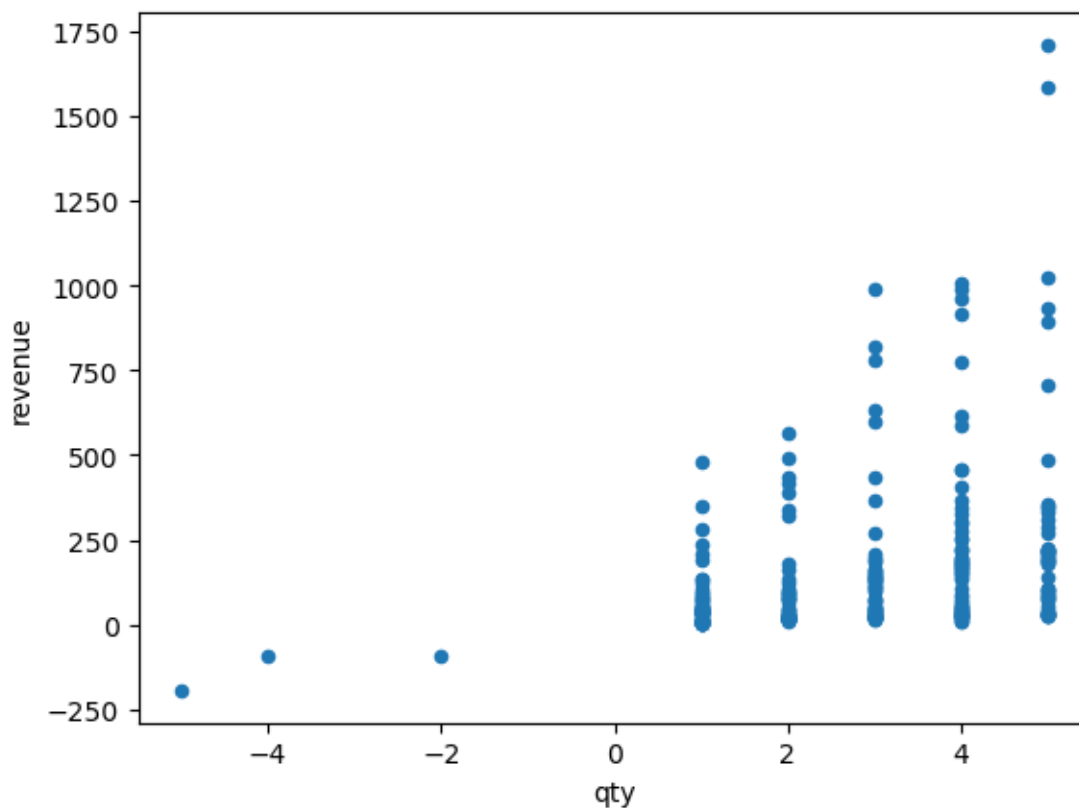
```
[11]:
```

	order_id	price	qty	zip	revenue
order_id	1.000000	-0.005070	0.024140	-0.014784	0.005711
price	-0.005070	1.000000	-0.023683	0.023812	0.817699
qty	0.024140	-0.023683	1.000000	0.074617	0.334196
zip	-0.014784	0.023812	0.074617	1.000000	0.044615
revenue	0.005711	0.817699	0.334196	0.044615	1.000000

Although step 6 says qty vs. revenue should be the standout, my results show that price vs. revenue has a stronger correlation. Qty vs. revenue has a Pearson's $r = 0.3342$, while price vs. revenue is 0.8177.

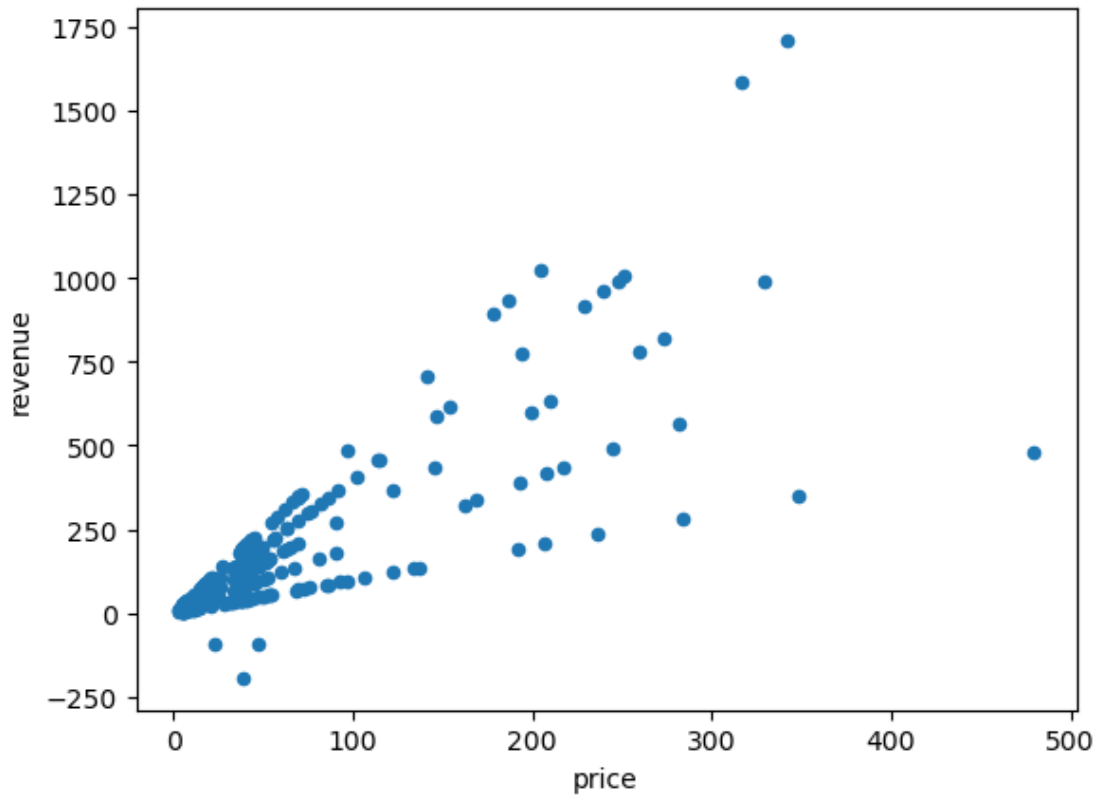
```
[12]: df.plot.scatter(x="qty", y="revenue")
```

```
[12]: <Axes: xlabel='qty', ylabel='revenue'>
```



```
[13]: df.plot.scatter(x="price", y="revenue")
```

```
[13]: <Axes: xlabel='price', ylabel='revenue'>
```



The quantity vs revenue scatter plot as an upward-drifting cloud, indicating a weak positive relationship. However the points are widely scattered, which supports the Pearson's r of 0.3342. The price vs revenue scatter plot shows a stronger positive relationship, which supports the Pearson's r of 0.8177. One caution is that a few high-priced products with very high revenue can influence the correlation.

0.0.1 By-hand check

[2, 4, 4, 6, 9]

Mean = $(2+4+4+6+9)/5 = 25/5 = 5$

Median = 4, the middle number is 4

```
[14]: # Created a pandas Series to verify my calculations
num = pd.Series([2, 4, 4, 6, 9])
print(num.mean())
print(num.median())
```

5.0

4.0

Git Hub Link: <https://github.com/jazzminowens/cs82a-portfolio>